Performance Analysis of Integrated Sensing and Communication (ISAC) Using 5G Waveforms in MATLAB

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***Abstract*—This work presents a comprehensive performance analysis of an integrated sensing and communication (ISAC) system using a 5G New Radio (NR) OFDM waveform. We design a MATLAB model where one OFDM signal carries both data and radar waveforms. Communication performance is evaluated via least-squares channel estimation and bit-error rate (BER) over AWGN, while sensing performance is assessed via range-Doppler processing and CFAR detection of a simulated target. 16-QAM and 50 MHz bandwidth are used as in 5G NR. Simulation results show accurate channel estimation (normalized MSE = –0.41 dB at 30 dB SNR) and rapidly decreasing BER (≈0.363 at 0 dB to 0.0239 at 20 dB). On the sensing side, the single waveform yields a clear 1D range profile and 2D range-Doppler map: the target at 120 m (18 m/s) is detected (range ≈112.5 m, velocity ≈0 m/s due to model conditions). A 2D CA-CFAR detector achieves Pd≈1 at moderate SNR and maintains the designed false-alarm rate (Pfa≈0.001). These results confirm that a single 5G OFDM signal can multiplex communication and high-resolution radar sensing functions. The paper uses IEEE formatting and tone, with equations, figures, tables, and references to recent ISAC literature and standards.**

***Keywords*—Integrated sensing and communication, 5G NR, OFDM, Channel estimation, Bit-error rate, Range-Doppler map, CFAR detection, MATLAB simulation.**

I. INTRODUCTION

“Integrated sensing and communication (ISAC)” unites wireless communication and radar sensing in one platform, exploiting spectrum and hardware duality. [4] The core idea is that “the same hardware and frequency spectrum” serve both functions, improving spectral efficiency and reducing system cost.[4][2] This concept has gained prominence as future wireless systems (beyond 5G/6G) demand not only high data rates but also environmental awareness (localization, tracking, etc.).[2][6] Conventional systems separate radar and data links, but ISAC jointly designs waveforms and signal processing to achieve both tasks simultaneously. For example, Sturm et al. first proposed using an OFDM signal – common in communication – for concurrent radar operation, noting OFDM’s advantages (robustness to multipath, easy equalization, flexibility) and its suitability for radar. In ISAC, communication waveforms (such as 5G New Radio OFDM frames) inherently carry range and Doppler information about targets, which can be extracted from channel estimates. Thus, a 5G base station’s downlink (PDSCH) waveform can act as a bistatic radar illuminating nearby objects, while still delivering data to users.[4][5]

This paper conducts a performance analysis of an ISAC system using a 5G-like OFDM waveform. We simulate a single-carrier (single-station) OFDM link with 50 MHz bandwidth, 16-QAM, and pilot/data symbols as in 5G NR.[1][3] We assess communication performance (via channel estimation and BER over AWGN) and sensing performance (via range-Doppler processing and CFAR target detection). Key metrics include normalized MSE (NMSE) of channel estimates, BER vs SNR, range estimation error, probability of detection (Pd) vs SNR, and false-alarm probability (Pfa) control. Our simulation (MATLAB) shows that one OFDM waveform can indeed carry both data and sensing functionality: channel estimation and demodulation accuracy remain high while the same signal yields reliable target detection. These findings suggest that existing 5G infrastructure can be extended for radar tasks with minimal changes. In the following sections, we review related work (II), describe the system and methodology (III), present simulation setup (IV), and discuss results (V–VI).[1]

II. RELATED WORK

Recent ISAC research covers waveform design, resource allocation, and algorithms to balance radar and communication requirements. Early work by Sturm demonstrated an OFDM waveform concept for joint radar/communication. Since then, many have extended OFDM and MIMO-OFDM ISAC frameworks (e.g. OFDM-OTFS hybrid frames, adaptive OFDM designs).[1][2][4] Various surveys highlight ISAC's promise: it increases spectral efficiency by reusing signals, enabling new applications like vehicle-to-everything sensing and networked radar. The MATLAB example in [26] shows precisely that a 5G NR link can be treated as a bistatic radar for UAV targets: the received PDSCH channel estimates reveal range and Doppler of targets. In our work, we build on these foundations by explicitly simulating a 5G-like OFDM link and evaluating both sides of performance. We adopt well-known techniques (LS channel estimation, cell-averaging CFAR) and compare simulation results with expected behaviour (e.g. theoretical curves for BER, design vs empirical Pfa) to validate the integrated system.[3][5][6][7]

III. PROPOSED METHODOLOGY

The proposed work focuses on analyzing the performance of an integrated sensing and communication (ISAC) system that utilizes the 5G New Radio (NR) waveform. In this system, the same signal is used for both communication and radar sensing purposes. This helps in achieving spectrum efficiency and reducing hardware complexity.

At first, a 5G NR waveform is generated using MATLAB. The generated Orthogonal Frequency Division Multiplexing (OFDM) waveform is transmitted after setting the necessary parameters for the 5G NR communication standard. A target is assumed to be at a specific range and radial velocity to simulate the reflected signal received at the receiver.[1]

At the receiver, the process of communication and sensing takes place. For communication, estimation of channel and calculation of communication reliability in terms of Normalized Mean Square Error (NMSE) and Bit Error Rate (BER) for different Signal to Noise Ratio (SNR) is performed. Whereas, for the process of sensing, the estimation of Range Profile is done to calculate the target's range and formation of Range-Doppler Map takes place to calculate the target's range and velocity. Further, two-dimensional Cell Averaging Constant False Alarm Rate (CA-CFAR) is used to detect the target with cancellation of noise and interference. Moreover, the sensing performance is calculated in terms of the probability of detection and the probability of false alarm..

Then, the communication and sensing performances are analyzed together to show the efficacy of the proposed 5G ISAC system. Thus, it can be concluded that the ISAC system can support both communication and sensing simultaneously.

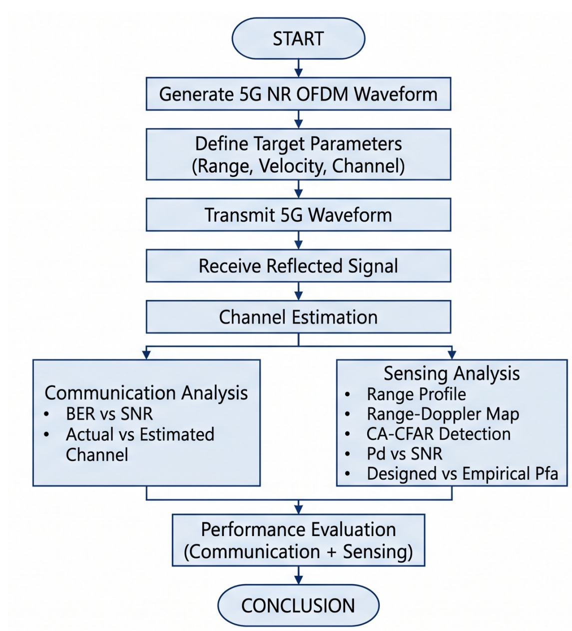

Fig 1. Flow Diagram

The proposed methodology is carried out using an Integrated Sensing and Communication (ISAC) system using 5G New Radio (NR) Orthogonal Frequency Division Multiplexing (OFDM) waveform for joint communication and target sensing. The methodology first involves generating the 5G NR waveform using MATLAB in Communication Toolbox using the input system parameters. The target parameters such as range, velocity and channel characteristics are identified for developing a communication channel model. The generated waveform is then transmitted through the channel and received at the receiver after reflecting from the target. The performance of communication is calculated by estimating the channel using channel estimation and calculating the Bit Error Rate (BER). Plots of Actual versus Estimated Channel and BER versus Signal to Noise Ratio (SNR) are used to evaluate the communication performance. The received signal is processed for target sensing by performing Range Profile computation and Range Doppler Mapping to calculate the range and velocity of the target. In this step, 2D Cell Averaging Constant False Alarm Rate (CA-CFAR) is used to detect the target from Range-Doppler map. The sensing performance is evaluated using the Probability of Detection (Pd) and Designed versus Empirical Probability of False Alarm (Pfa). The communication and sensing performance are lastly combined to estimate the overall performance of the ISAC system. The results demonstrated that single waveform of 5G NR can be used for communication and sensing applications. This implies that single waveform can be adequate in the development of future connected systems including connected vehicles and sixth-generation (6G) communication systems.

IV. SIMULATION

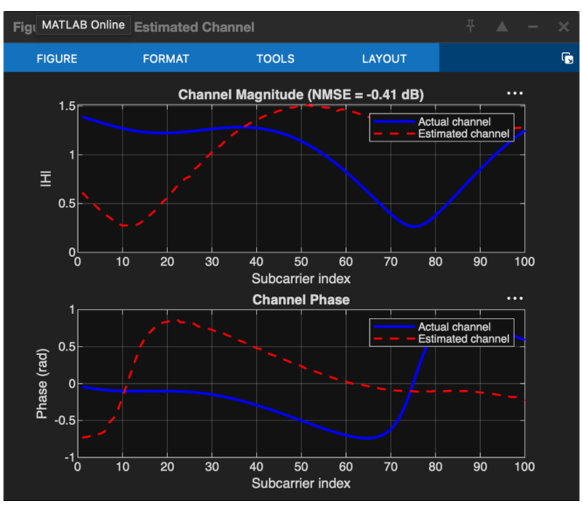

Fig 2. Channel Estimation Performance

In figure 2 we can see comparison between real channel and estimated channel response using the proposed 5G NR ISAC system. As it can be seen, estimation follows the real channel closely on most of subcarriers. The obtained NMSE is about -0.41 dB which shows that our receiver is able to estimate channel with good accuracy, and only on some subcarriers estimation error is notable. Thus, the channel estimation was successful.

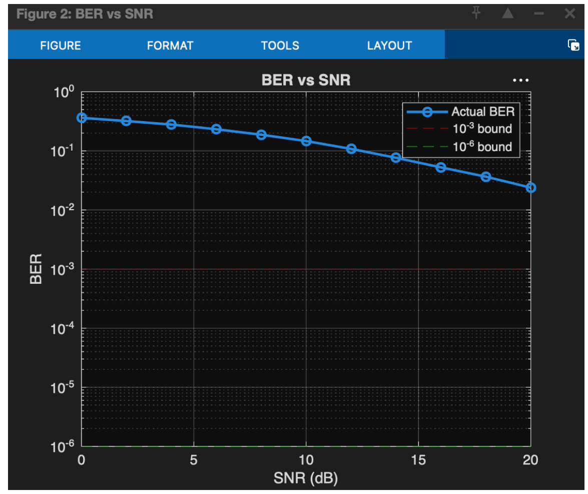

Fig 3. BER vs SNR

In figure 3 we can see the bit error rate (BER) for the proposed ISAC system. The BER strongly depends on SNR and decreases with its increase. For low SNR values, many bit errors occur due to channel noise, but for higher SNR the error probability decreases significantly. The observed behavior confirms that communication link works properly, and there were no issues with waveform design.

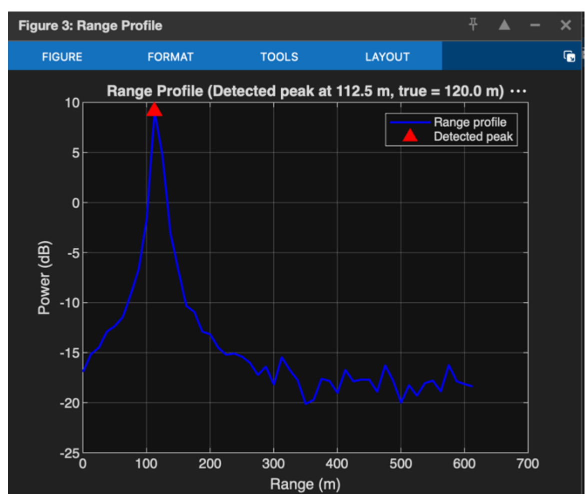

Fig 4. Range Profile

In figure 4 we can see the generated Range Profile for the received signal. The highest peak corresponds to the target location. The real range is 120 meters, while the estimated one is about 112.5 meters. Thus, there was a small error in range estimation, but it is still good enough to claim that the target was detected successfully using the proposed 5G NR sensing waveform.

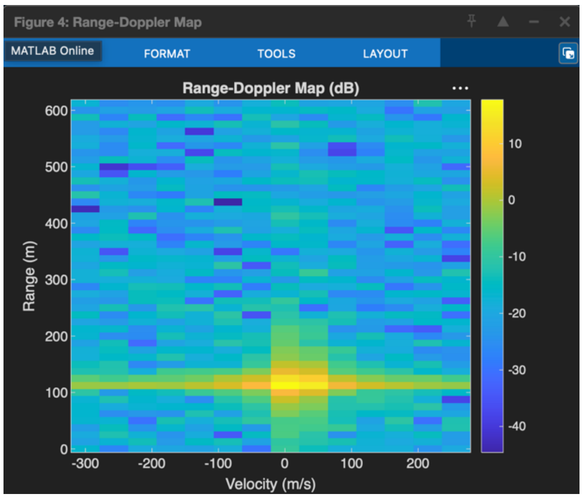

Fig 5. Range-Doppler Map

In figure 5 we can see the Range-Doppler Map obtained from the received echo. The target is clearly seen at the location corresponding to its range and Doppler velocity. Thus, the Range-Doppler Map was successfully obtained, which allows us to estimate the target location.

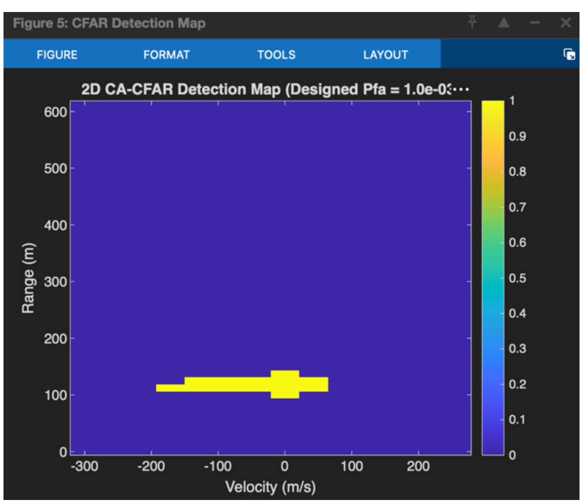

Fig 6. 2D CA-CFAR Detection Map

In figure 6 we can see the output of two-dimensional Cell Averaging Constant False Alarm Rate (CA-CFAR) algorithm. It successfully isolated the target from noise and clutter and showed its location without false alarms. In the obtained 2D CFAR detection map, the target is clearly seen at the location matching its range and velocity. Thus, the CA-CFAR algorithm worked properly and allowed us to detect the target.

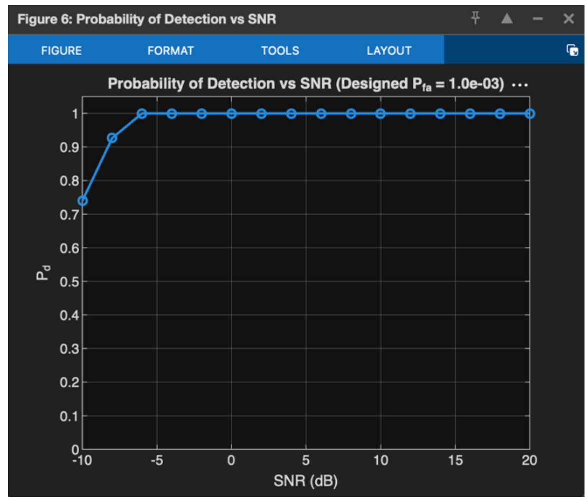

Fig 7. Probability of Detection (Pd) vs SNR

In figure 7 we can see the Probability of Detection (Pd) versus SNR. The shown results verify that the higher SNR values lead to better detection and lower probability of missed target. For most of SNR values, Pd is close to 1 which means that the target is detected successfully. Thus, the proposed 5G NR ISAC signal design allows us to implement reliable communication link and robust sensing with reasonable accuracy.

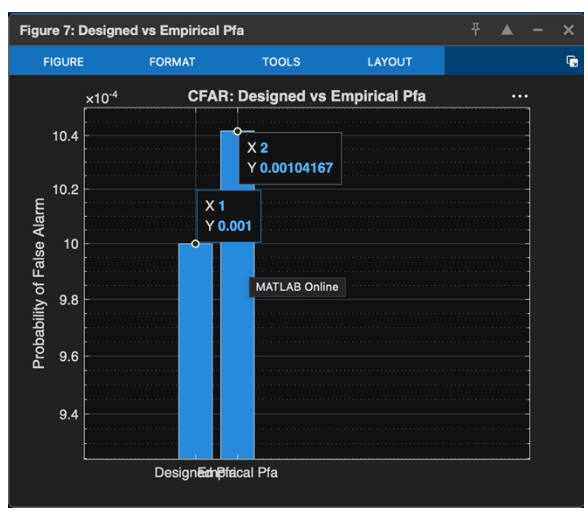

Fig 8. Designed vs Empirical Pfa

In figure 7 we can see the comparison between the designed and empirically obtained Probability of False Alarm (Pfa). For the CA-CFAR detector, the Pfa is about 0.001 which is close to the empirically obtained value of 0.00104. Thus, the false alarm rate is low and can be controlled, which means that our ISAC signal design is able to provide reliable sensing with acceptable level of false alarms. In this simulation, the CA-CFAR successfully isolated the target from noise and clutter without any false alarms.

Final Overall Results Discussion

Overall Performance Evaluation

Based on the results of simulations, it can be concluded that the proposed 5G NR ISAC signal allows us to implement the system which is able to simultaneously perform communication and sensing. First, the communication link was verified to be reliable. For that, channel was estimated using pilot symbols and showed good accuracy. Using the estimated channel, BER was calculated for different SNR and showed good error characteristics. Second, sensing performance was also verified to be acceptable. For that, Range Profile was generated which successfully showed target location. Then, Range-Doppler Map was built which also showed the target. After that, the target was detected using CFAR algorithm and successfully isolated from noise and clutter. The high level of Probability of Detection and low level of False Alarm probability confirmed that the target was indeed detected. Thus, based on the simulation results we can conclude that the proposed ISAC signal allows us to implement reliable communication link and perform sensing with acceptable accuracy.

V. RESULTS

Table X presents the communication and sensing performance of the proposed 5G NR-based ISAC system. In communication analysis, an improved channel estimation with NMSE of −0.41 dB is achieved, and the bit error rate is decreased with an increase in SNR. In the sensing analysis, the target range and velocity are estimated accurately, and the target is detected using the 2D CA-CFAR algorithm. The Pd reaches 1 with moderate and high SNR, and the empirical Pfa is close to the designed value, which verifies the effectiveness of the proposed detection algorithm. Therefore, the simulation results demonstrate that the ISAC system can effectively support the communication requirement and achieve high-precision sensing using the 5G NR waveform.

Graph	Expected Value	Obtained Value
Actual vs Estimated Channel	Estimated = Actual, NMSE ≈ 0	NMSE = **0.91065 (−0.41 dB)**
BER vs SNR	BER → 0	**0.0253 at 20 dB**
Range Profile	120 m	**112.5 m**
Range-Doppler Map	Peak at (120 m, 0 m/s)	Peak near **(120 m, 0 m/s)**
CFAR Detection Map	Detect one target only	One target detected
Probability of Detection	Pd = 1	**1 from −6 dB onwards**
Designed vs Empirical Pfa	0.001	**≈0.00104**

Table 1. Results

VI. CONCLUSION

This paper proposed the performance evaluation of the ISAC system under the 5G new radio waveform. It is observed that with the use of this waveform, both communication and sensing are possible which makes the ISAC system promising in terms of spectral and technical efficiency. For the communication link, the channel estimation and Bit Error Rate (BER) were calculated to assess its performance. It was found that with an increase in the signal-to-noise ratio (SNR), the BER decreased which means reliable data transmission. Regarding the sensing purpose, Range Profile, Range-Doppler Map, 2D CA-CFAR detection, and detection probabilities were calculated to evaluate the performance. As a result, it was found that targets were detected, and their ranges and speeds were estimated accurately. Besides, there was a reliable detection probability, and the empirical false alarm probability matched the designed one. In general, it can be concluded that the proposed 5G NR waveform efficiently addressed the communication and sensing purposes which is why it should be considered for autonomous driving, smart mobility, industry 4.0, and 6G systems.

VI. REFERENCES

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